



Effects of Mach number and 3D instabilities on boundary-layer transition over gaps

Víctor Ballester Ribó¹, Jeffrey Crouch²,
Yongyun Hwang¹, Spencer Sherwin¹

¹Department of Aeronautics, Imperial College London, UK

²The Boeing Company, USA

29 July 2026

IMPERIAL



Motivation

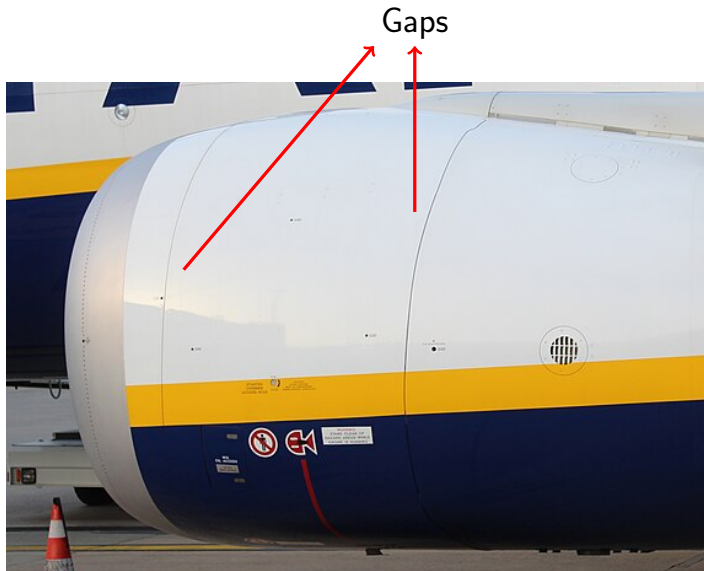


Figure: Engine nacelle of a Boeing 737

First questions

- Small disturbances in the boundary layer initially grow as $\propto e^{n(x)}$.
- We aim to understand how much a gap of width w and depth d can overamplify these disturbances.
- For each gap of width w and depth d , we define

$$\Delta n_{\text{gap}} := n_{\text{gap}} - n_{\text{flat}}$$

Experimental observations

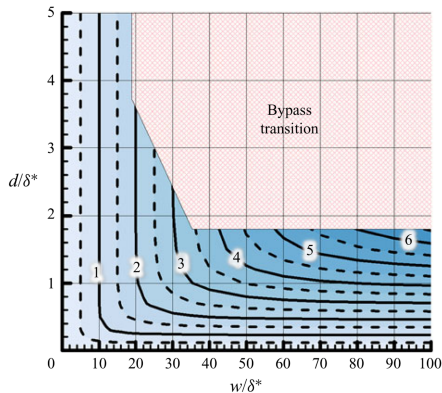
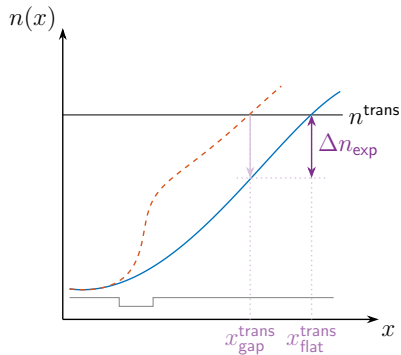


Figure: Δn_{gap} contours in the w - d parameter space, from experiments by Crouch et al. (2022)[†].

[†] Crouch JD, Kosorygin VS, Sutanto MI, Miller GD. Characterizing surface-gap effects on boundary-layer transition dominated by Tollmien–Schlichting instability. *Flow*. 2022;2:E8.

Numerical setup

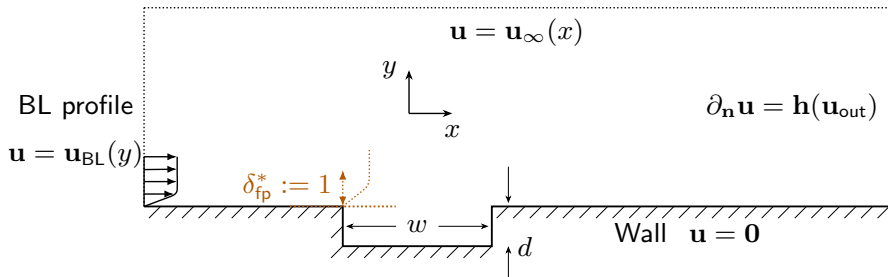


Figure: Domain setup for the steady-state finder

Aims:

- Study the stability of the system as a function of the depth d and width w of the gap. Reynolds number is fixed to

$$\text{Re} = \frac{u_{\infty} \delta_{fp}^*}{\nu} = 1000.$$

- Understand the influence of the Mach number.
- To what extent are 3D effects important?

2d incompressible limit set classification

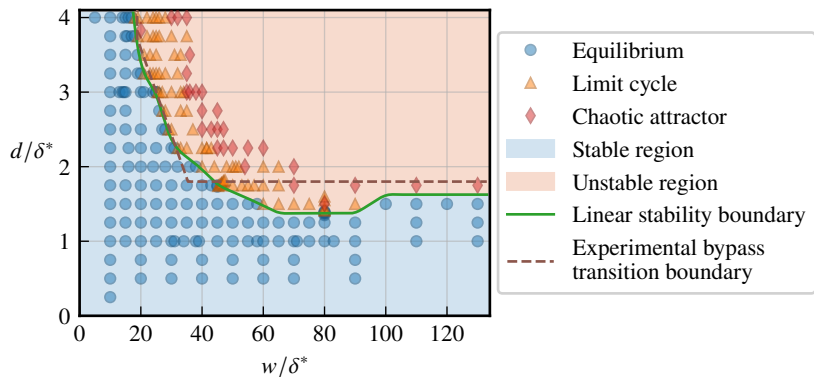
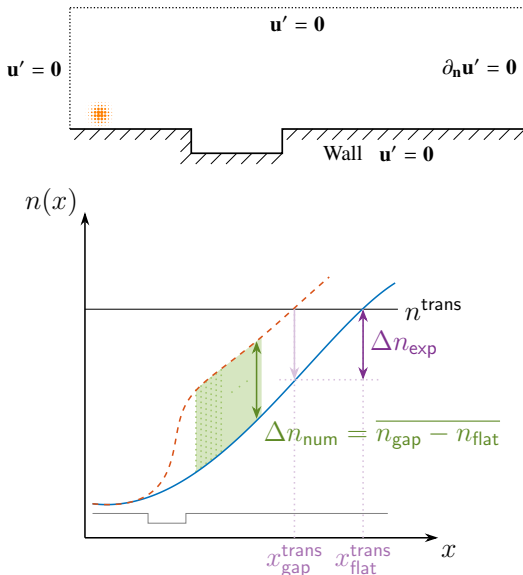


Figure: Limit set classification over the parameter space $10 \leq w/\delta^* \leq 130$ and $0.25 \leq d/\delta^* \leq 4$.

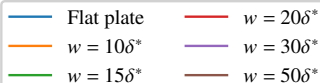
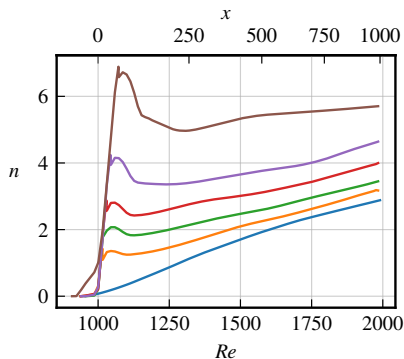
Stability setup to excite the boundary layer

- We linearize the flow around a steady base flow.
- We add a stochastic forcing term in the upstream part of the boundary layer as a form of white Gaussian noise.

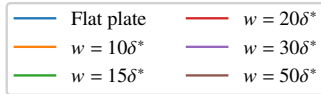
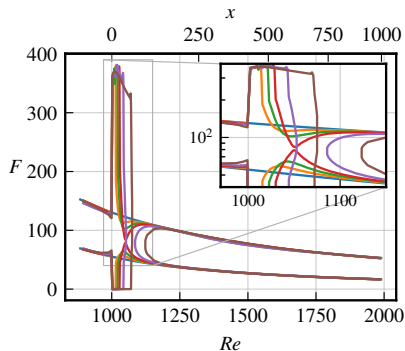


n -factor curves

Evolution of n for disturbances growing as $\propto e^{n(x)}$



Spatial stability curves for the parallel flow assumption



Numerical predictions of Δn

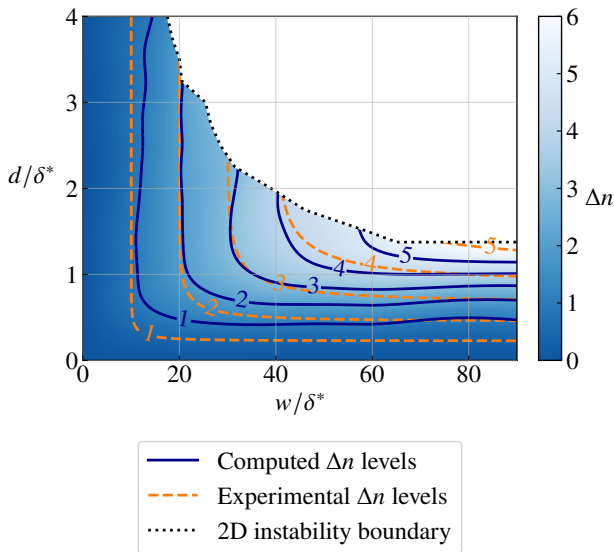


Figure: Contours of Δn in the w - d parameter space.

3d effects

- We perform single mode stability analysis to find the largest growth rate of 3D modes as a function of the spanwise wavenumber β .

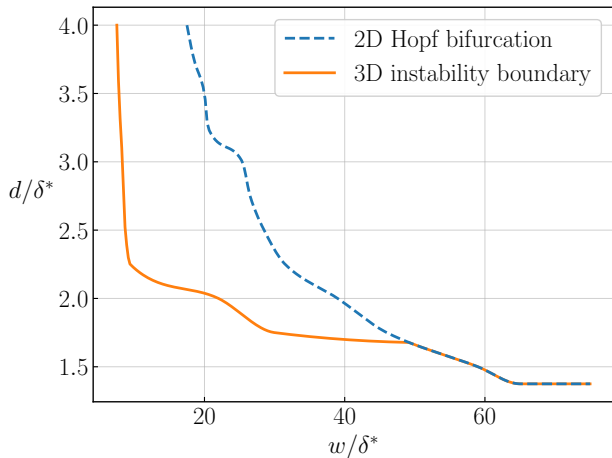
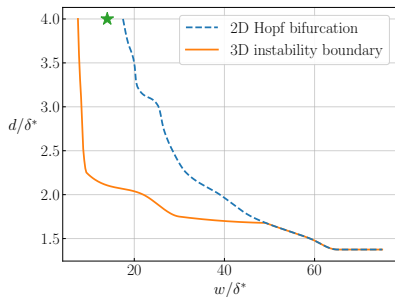


Figure: Bifurcation boundaries for the 2D and 3D instabilities.

3d non-linear results



- Long streaky structures appear in the downstream part of the gap
- Chaotic behaviour is observed in the gap, but insufficient to destabilise the system.

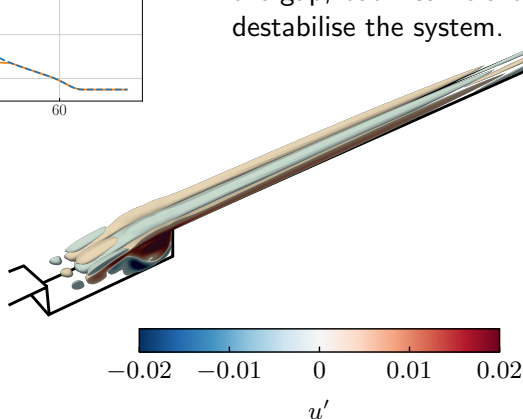
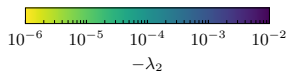
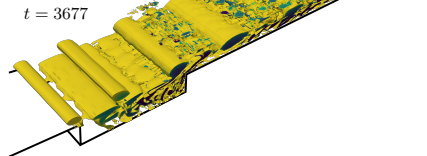
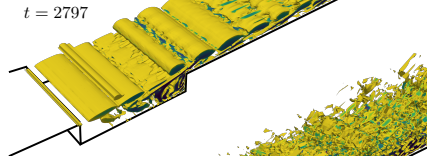
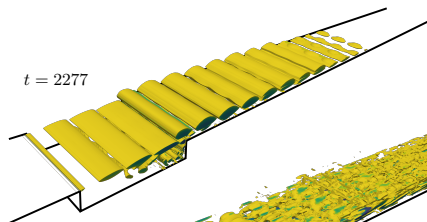
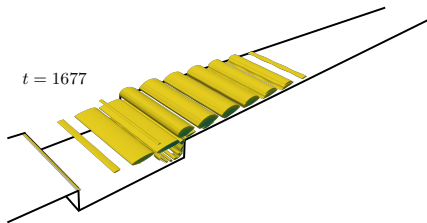
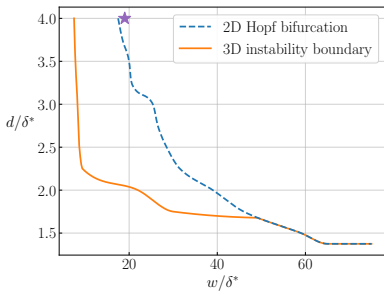


Figure: Most unstable mode for the 3D instability.

3d transition



Mach number effects

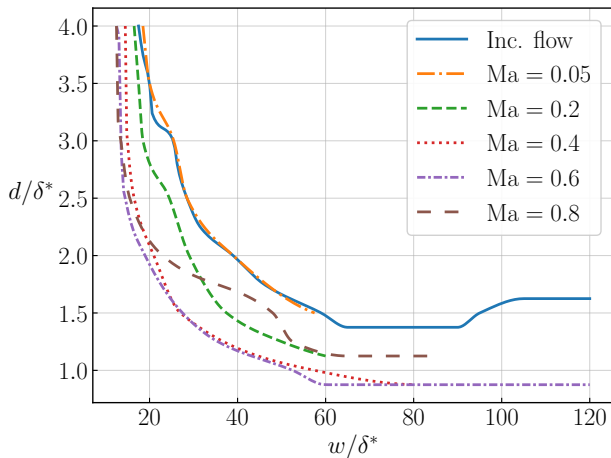


Figure: Hopf bifurcation boundaries for different Mach numbers.

Conclusions

- Very good agreement between the numerical predictions (2d inc.) and the experimental (3d low Mach wind tunnel) results.
- The 3D instability doesn't seem to be strong enough to destabilise the downstream boundary layer. Only after the 2D instability sets in, the transition location advances till the downstream edge of the gap.
- For deep gaps, Mach number has a destabilising effect, while for shallow gaps, higher Mach numbers lead to later transition.